M4 · SMEC Module 4, manufacturing processes, thermal energy and mechanisms

Manufacturing processes, energy conversions & mechanisms

Sand casting, metal forming, welding polarities, metal cutting and CNC, the 7 ASTM additive manufacturing categories, IC engines, power plants, refrigeration, and kinematic linkages.

○ Mark as done ○ Highlight Quiz this module · FAT PYQs

All syllabus sections & lecture slide diagrams included · Ticks itself when every tab is read

Casting & Forming ✓ | Welding & CNC | 3D Printing (ASTM) | IC Engines (Otto/Diesel) | Power Plants & RAC | Mechanisms & Hexapod

Syllabus Order: 1. Casting • 2. Forming • 3. Joining • 4. Machining & CNC • 5. 3D Print • 6. IC Engine • 7. Power Plants • 8. RAC • 9. Fuel Cell • 10–12. Mechanisms & Robotics

Sources: SMEC VIT Module 3 & 4 Full Notes • VIT SMEC Question Bank

M4T1 Module 4, topic 1

Basic Metal Casting (Green Sand Mold Anatomy, Allowances & Defects)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context

Metal Casting is the oldest and most versatile manufacturing route: melt metal, pour it into a refractory sand mold cavity, let it freeze, and break the sand away (SMEC VIT Module 4 Notes). Memorize the labeled **Green Sand Mold Cross-Section in Figure 4.1 (Left)**—it is one of the highest-frequency 10-mark diagrams in VIT FAT exams!

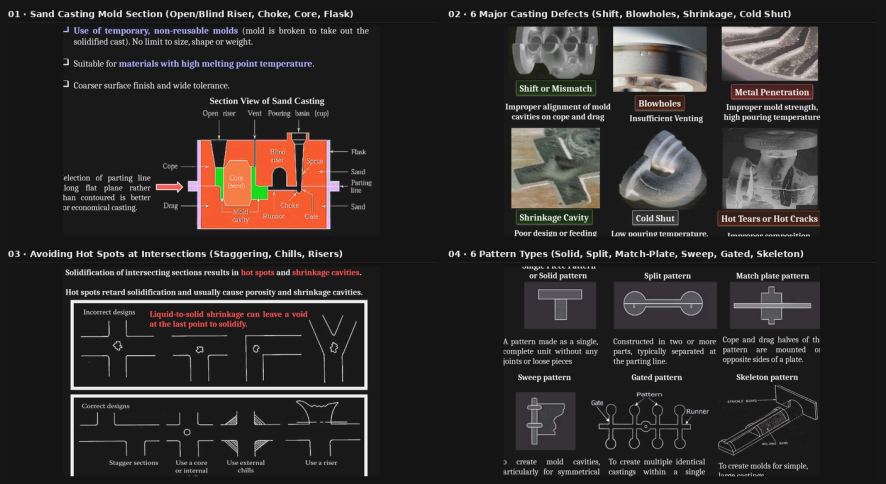


Figure 4.1B (Extracted from VIT Lecture 11 Slides): (a) Sand Casting Section View (Open/Blind Riser, Choke, Core), (b) 6 Major Casting Defects (Shift, Blowholes, Penetration, Shrinkage, Cold Shut, Hot Tears), (c) Avoiding Hot Spots at Intersections (Staggering, Chills, Risers), (d) 6 Pattern Types.

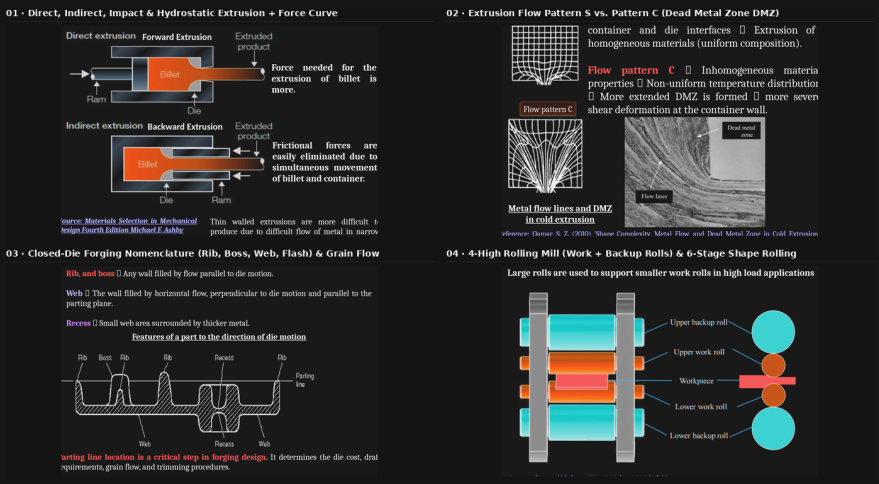


Figure 4.1C (Extracted from VIT Lecture 12 Slides): (a) Direct, Indirect, Impact & Hydrostatic Extrusion + Force vs. Ram Displacement Graph, (b) Extrusion Flow Pattern S vs. Pattern C (Dead Metal Zone), (c) Closed-Die Forging Nomenclature (Rib, Boss, Web) & Grain Flow, (d) 4-High Rolling Mill (Work + Backup Rolls) & Shape Rolling.

Complete Lecture 11

Additional Sand Mold Terminology from Slides	Besides Cope (top flask), Drag (bottom flask), and Cheek (intermediate flask for 3-part molds), the gating system includes the **Choke** (the constriction at the base of the sprue that controls molten metal flow rate and traps slag in the pouring basin), **Open Riser** (open to atmosphere at top of cope), and **Blind Riser** (domed reservoir enclosed inside the sand next to the casting). *Rule of Parting Line & Riser Placement:* Selecting a flat parting line is always more economical than a contoured parting line; the **Riser must always be attached to the heaviest/thickest section** so the riser is the very last region to solidify (directional solidification).
6 Major Sand	**6 Major Sand Casting Defects & Exact Causes (High-Probability FAT Match/Short-Note Question!):** 1. **Shift (Mismatch):** Lateral step at parting line caused by misaligned pins/flasks between Cope and Drag. 2. **Blowholes:** Smooth spherical gas cavities caused by **insufficient mold venting** or low sand permeability trapping steam/gases. 3. **Metal Penetration:** Rough, sandy surface finish caused by coarse sand grains, soft ramming, or **excessively high pouring temperature** allowing liquid metal to penetrate between sand grains. 4. **Shrinkage Cavity:** Jagged internal/surface void at thick sections caused by poor feeding or undersized/misplaced risers during liquid-to-solid volumetric contraction. 5. **Cold Shut:** Two streams of molten metal meet from different gates but fail to fuse into one continuous part due to **too low pouring temperature** or slow pouring. 6. **Hot Tears (Hot Cracks):** Jagged cracks occurring right after solidification when contraction of the hot weak casting is restrained by a rigid mold or core.
Hot Spots at	**Hot Spots at Intersecting Sections (T, X, L, Y Junctions) & 4 Engineering Remedies:** Intersecting walls have a smaller surface-area-to-volume ratio, retarding cooling and creating a localized **Hot Spot** that leaves a shrinkage void. *4 Slide Remedies:* (1) **Stagger intersecting sections** (replace an X-junction with two offset T-junctions), (2) **Insert a Core or Internal Chill** at the thick center to maintain uniform wall thickness, (3) **Place External Metal Chills** in the sand corners to accelerate local heat extraction, or (4) **Place a Riser** directly over the heavy junction.
6 Types of Patterns, Single vs. Double Draft & Chaplets	**6 Pattern Types:** (1) **Single-Piece (Solid) Pattern** (one piece, no joints, for simple flat backs), (2) **Split Pattern** (two-piece pattern split along parting plane with dowel pins for complex/symmetrical shafts or valves), (3) **Match-Plate Pattern** (cope and drag halves

mounted on opposite sides of a single metal/wooden plate for rapid machine molding), (4) **Sweep Pattern** (a contoured wooden board rotated around a vertical spindle to sweep large symmetrical bell/dome molds in green/loam sand), (5) **Gated Pattern** (multiple patterns permanently interconnected by runners and gates for mass-producing small castings in one pour), and (6) **Skeleton Pattern** (a ribbed wooden frame filled with loam sand and smoothed with a **strickle board** for huge cast-iron pipes or turbine casings where a solid wooden pattern would be too expensive).

Single-Draft vs. Double-Draft Bush & Chaplets: Molding a hollow bush half in the Cope and half in the Drag requires a **Double Draft Angle** (taper on both halves), whereas molding it entirely in the Drag with a flat Cope requires only a **Single Draft Angle**! **Chaplets** are small spool-shaped metal spacers placed inside the mold cavity to support a heavy sand core against sagging before pouring and against buoyancy floating after pouring; the molten metal fuses the chaplets permanently into the finished casting (so chaplets must be made of the *same alloy* as the casting!).

Complete Lecture 12

Extrusion Classifications & Force vs. Ram Displacement Graph

Extrudability Order: Carbon steels are difficult to extrude (high flow stress & high temperature causing die wear), whereas non-ferrous metals (**Aluminium, Copper, Magnesium, Zinc, Brass**) have high extrudability. Complexity is measured by the **Circumscribing-Circle Diameter (CCD)**—the smallest circle that encloses the cross-section (Al extrusions reach 1.02 m CCD; steels max ~150 mm). *Good Extrusion Design Rules:* Keep wall thicknesses uniform, avoid sharp inside corners (use generous fillets to prevent stress concentration), and avoid knife-edges or closed hollow sections unless extruded over a mandrel/spider die.

4 Extrusion Types: (1) **Direct (Forward) Extrusion** (billet slides along container wall in same direction as ram; **highest peak ram force** due to container wall friction, which drops steadily as billet shortens), (2) **Indirect (Backward) Extrusion** (hollow ram pushes die into stationary billet; billet does NOT move relative to container wall → **container wall friction is completely eliminated** and ram force stays flat and low!), (3) **Impact Extrusion** (high-speed punch strikes a cold slug to squirt thin-walled collapsible toothpaste tubes, aerosol cans, and battery shells upward around the punch), and (4) **Hydrostatic Extrusion** (billet is surrounded by a pressurized fluid medium so wall friction is zero and brittle materials can be extruded under hydrostatic compression).

Metal Flow Pattern S vs. Flow Pattern C: *Flow Pattern S* occurs in homogeneous, frictionless flow (well-lubricated or indirect extrusion). *Flow Pattern C* occurs when high container-wall friction or rapid cooling of the billet skin arrests the outer corners, creating a stagnant **Dead Metal Zone (DMZ)** ahead of the die and severe internal shear!

Closed-Die Forging Nomenclature, Grain Flow & Draft Allowances

Forging Nomenclature: Web = thin horizontal section formed perpendicular to die motion; **Rib & Boss** = vertical walls/projections filled by metal flowing parallel to die motion; **Flash** = excess metal squeezed into the narrow flash gutter at the parting line (throttles outward flow so high cavity pressure forces metal into deep ribs and corners, then trimmed off).

Grain Flow Superiority: In *Casting*, crystal grains are random and isotropic (lowest toughness); in *Machining from bar stock*, tool paths cut straight across grain lines (exposing grain ends to fatigue crack initiation and stress corrosion); in *Forging*, unbroken grain fibers deform continuously to follow the exact contour of the component (e.g., crank webs, connecting rods, spanners, gears)—giving maximum fatigue life and impact resistance!

Internal vs. External Forging Draft Angles: As a hot forging cools inside the die, it shrinks radially. Therefore, it pulls away from outer die walls (requiring a smaller **External Draft Angle of 3°–7°**) but shrinks tightly onto internal die projections/plugs (requiring a larger **Internal Draft Angle of 7°–10°** so it does not seize on the punch!).

Rolling (4-High Mill)

	Why a 4-High Rolling Mill Uses Small Work Rolls + Large Backup Rolls: Thin strip rolling requires a small-diameter **Work Roll** to minimize the contact arc length ($L = \sqrt{R \cdot \Delta t}$), which lowers the required rolling force and prevents roll flattening. Because slender work rolls would bend/flex under load (producing a crown/thick center), huge rigid **Backup Rolls** support the work rolls from above and below! **6-Stage Shape Rolling of an I-Beam / Railway Rail:** Square cast bloom passes sequentially through grooved rolls: **Blooming pass → Edging pass → Roughing pass → Intermediate pass → Edging pass → Finishing pass**. **Lecture 12 Slide 20 Part Identification Quiz (Exact Exam Questions!):** (1) **Automotive Alloy Wheel:** *Low-Pressure / High-Pressure Die Casting* (complex 3D spokes + high surface finish), (2) **Engine Connecting Rod:** *Closed-Die Drop Forging* (requires contoured unbroken grain flow to survive alternating cyclic tensile/compressive combustion loads), (3) **Railway I-Section Track:** *Hot Shape Rolling* (long continuous uniform profile with refined wrought grain structure), (4) **Aluminium Slotted T-Profile & Finned Heat Sink:** *Direct Hot Extrusion* (intricate constant cross-section with high CCD), and (5) **Ventilated Cast-Iron Brake Disc:** *Sand Casting (with core for internal cooling vanes) followed by CNC Turning, Facing & Drilling*!
Anatomy of a Sand Casting Mold (Figure 4.1-1)	
Molding Flask (Cope & Drag)	Rigid metal/wood frame holding the green sand (silica sand + 10% bentonite clay + 4% water). The **Cope** is the top half; the **Drag** is the bottom half; their meeting plane is the **Parting Line**.
Pattern & Its 3 Allowances	A wood/metal replica of the final part used to pack sand around. Why is a pattern slightly larger than the finished part? Because of (i) **Shrinkage Allowance** (metal contracts as it cools from liquid to room temp), (ii) **Draft / Taper Allowance (1°–3° vertical taper)** so the pattern pulls out of the sand without tearing the mold walls, and (iii) **Machining Allowance** (extra 2–3 mm layer to be machined smooth later).
Core, Core Prints & Chaplets	A **Core** is a baked dry-sand insert placed inside the cavity (resting on recesses called **Core Prints**) to create **hollow internal cavities or holes**. **Chaplets** are small metal spacers (made of the same metal as the casting) placed above and below the core to stop it from sagging under gravity or floating upward by buoyancy when molten metal rushes in—they melt and fuse permanently into the casting wall!
Gating System & Riser	Molten metal flows through **[Pouring Basin] → [Sprue (tapered vertical channel to prevent air aspiration)] → [Runner (horizontal channel in Drag)] → [Ingate] → [Mold Cavity]**. The **Riser (Feeder)** is an extra vertical reservoir connected to the thickest section that stays liquid longest and feeds molten metal back into the casting as it shrinks during solidification!
6 Classic Casting Defects	(1) **Misrun:** Metal solidifies before completely filling the cavity (pouring temp too low). (2) **Cold Shut:** Two streams of liquid metal meet inside the mold but fail to fuse, leaving a weak seam. (3) **Blowholes / Pinholes:** Smooth round gas cavities caused by trapped steam/gases (poor sand permeability or venting). (4) **Shrinkage Cavity:** Jagged internal void caused by an undersized Riser. (5) **Mold Shift:** Misalignment of Cope and Drag at the parting line. (6) **Hot Tears:** Cracks caused by restrained contraction right after solidification.

M4T2 Module 4, topic 2

Metal Forming Processes (Hot vs. Cold Working, Rolling, Forging, Extrusion)

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context	In **Metal Forming**, zero metal is melted or wasted as chips—instead, we apply massive forces exceeding the metal's *Yield Strength* to plastically deform it into shape (**Figure 4.1-2**).
Hot Working (>)	**Hot Working (> Recrystallization Temp, 0.5–0.75 T_m) vs. Cold Working (Room Temp < T_R):** In *Hot Working*, new strain-free crystal grains form continuously—requiring low force and allowing huge shape changes without cracking, but leaving a rough oxidized surface scale. In *Cold Working*, dislocations tangle up (**Strain Hardening / Work Hardening**), requiring high force and reducing ductility, but delivering **higher tensile strength, mirror surface finish, and tight dimensional tolerances**!
Rolling (Figure 4.1-2A)	Two counter-rotating cylindrical rolls pull a slab in by friction and squeeze it thinner (produces sheets, plates, railway tracks, I-beams).
Forging (Figure 4.1-2B)	Squeezing or hammering metal between dies. Because metal crystal grains flow continuously along the contour of the part (**unbroken grain flow**), forged parts have the highest impact and fatigue strength (used for IC engine crankshafts, connecting rods, spanners, landing gear). *Open-Die Forging* uses flat dies for giant shafts; *Closed-Die (Impression) Forging* squeezes metal into a precision cavity where excess metal escapes into a thin gutter called **Flash**.
Extrusion vs. Wire Drawing (Figure 4.1-2C & D)	In **Extrusion**, a hydraulic ram **PUSHES** a heated billet through a shaped die opening like squeezing toothpaste (used for complex hollow Aluminium window frames). In **Wire / Rod Drawing**, a tensile force **PULLS** a pointed rod through a tapered carbide die to reduce its diameter (used for electrical copper wires and steel cables).

M4T3 Module 4, topic 3

Metal Joining Processes (Arc Welding: SMAW, TIG, MIG & Soldering)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	**Welding** permanently fuses two base metals together using heat and/or pressure. Look at the **3 Metallurgical Zones of a Fusion Weld** and the **TIG vs. MIG Setups** in **Figure 4.2 (Panel 1)** (SMEC VIT Module 4 Notes):

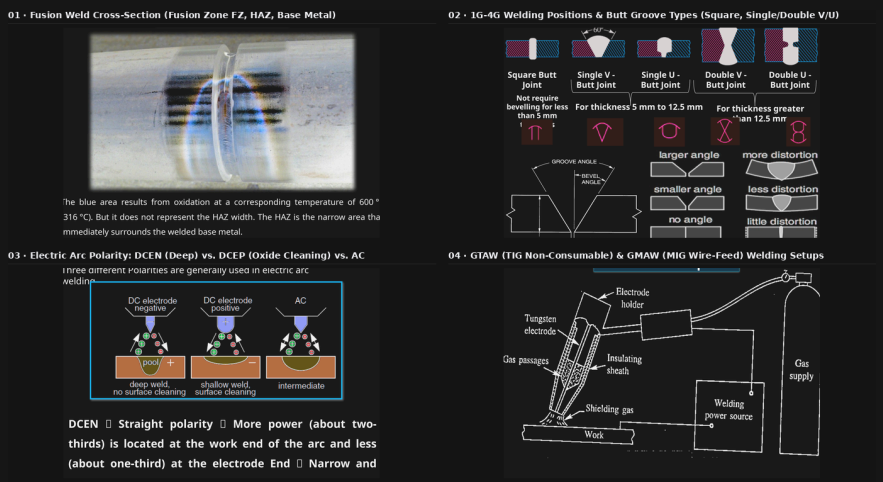


Figure 4.2B (Extracted from VIT Lecture 13 Slides): (a) Weld Cross-Section (Fusion Zone FZ, HAZ, Base Metal), (b) 1G–4G Positions & Butt Groove Types (Square, Single-V/U, Double-V/U), (c) Arc Polarity (DCEN Deep Penetration vs. DCEP Oxide Cleaning vs. AC), (d) GTAW (TIG) & GMAW (MIG) Setups.

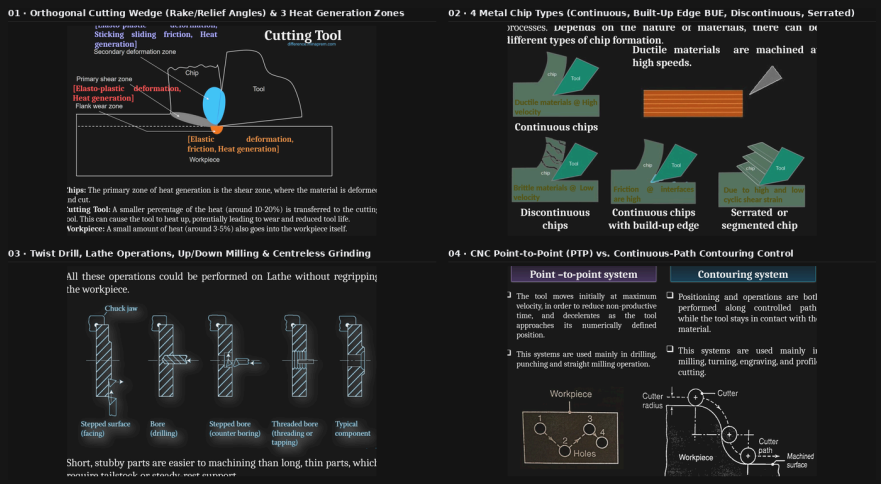


Figure 4.2C (Extracted from VIT Lecture 14 Slides): (a) Orthogonal Cutting Geometry (Rake/Relief Angles) & 3 Heat Zones, (b) 4 Chip Types (Continuous, BUE, Discontinuous, Serrated), (c) Twist Drill, Lathe, Milling & Centreless Grinding, (d) CNC Point-to-Point vs. Contouring Control.

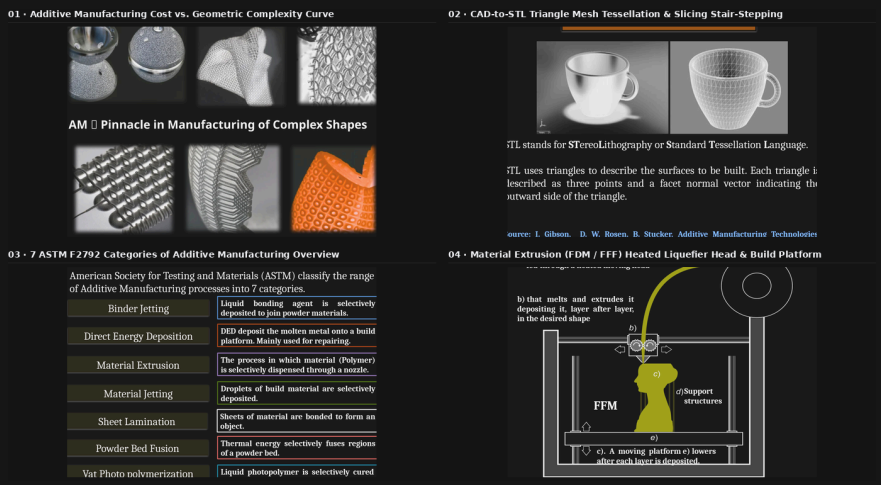


Figure 4.2D (Extracted from VIT Lecture 15 Slides): (a) AM Cost vs. Complexity Curve, (b) CAD-to-STL Triangle Tessellation & Slicing Stair-Step Effect, (c) 7 ASTM Categories of Additive Manufacturing, (d) FDM Extruder Head & Build Platform.

Complete Lecture 13

Weld Zones & Why Fracture Occurs in the HAZ

Every fusion weld exhibits three distinct metallurgical zones: (1) **Fusion Zone (FZ / Weld Metal)** where filler and base metal melted and solidified into columnar dendrites, (2) **Heat Affected Zone (HAZ)**—the unmelted solid base metal immediately adjacent to the fusion line that experienced peak temperatures high enough to cause grain coarsening, recrystallization, or tempering/softening (making the **HAZ the weakest zone where tensile failure typically initiates!**), and (3) unaffected **Base Metal (BM)**.

Power Density Spectrum

Gas Welding (Oxyacetylene) has the lowest power density (~**10 W/cm^2**) → slow heating, wide saucer-shaped FZ, wide HAZ, and high thermal distortion; **Electric Arc Welding (SMAW, TIG, MIG, PAW, SAW)** operates around 10^3–10^4 W/cm^2; **Power Beam Welding (Electron Beam EBW & Laser Beam LBW)** reaches **10,000+ W/cm^2** → deep, needle-narrow keyhole weld with almost zero HAZ and minimal distortion!

Welding Positions (1G–4G) & Plate-Thickness Butt Groove Selection

Positions: 1G = Flat (easiest, highest deposition rate), **2G = Horizontal**, **3G = Vertical**, **4G = Overhead** (hardest; molten puddle fights gravity).

Butt Groove Rules by Plate Thickness t: Square Butt Groove for thin plates (**t < 5 mm**); **Single-V or Single-U Groove** for medium plates (**t = 5 to 12.5 mm**, 60°–70° bevel angle); **Double-V or Double-U Groove** for thick plates (**t > 12.5 mm**—requires half the weld metal of a Single-V and balances shrinkage stresses on both sides to prevent angular warpage!).

3 Electric Arc	**3 Electric Arc Polarities (DCEN vs. DCEP vs. AC — Classic FAT Conceptual Question!):** **DCEN (Direct Current Electrode Negative — "Straight Polarity"):** Workpiece is (+), Electrode is (−). Electrons accelerate from electrode and slam into the workpiece, releasing **2/3 (67%) of arc heat at the workpiece** and only 1/3 at the electrode → gives a **deep, narrow weld penetration** and allows high current on a small tungsten electrode without melting it! **DCEP (Direct Current Electrode Positive — "Reverse Polarity"):** Workpiece is (−), Electrode is (+). Electrons slam into the electrode (2/3 heat at electrode, requiring a larger water-cooled tungsten tip), resulting in a **wide, shallow weld**. Crucially, heavy positive inert gas ions (**Ar^+**) bombard the negatively charged workpiece surface like sandblasting, knocking off stubborn, high-melting-point refractory oxide films (**Al_2O_3 on Aluminium and MgO on Magnesium**)—known as the **Cathodic Oxide Cleaning Action**! **AC (Alternating Current):** Alternates between DCEN (deep heating half-cycle) and DCEP (oxide-cleaning half-cycle), giving symmetric 50%/50% heat distribution—the standard choice for **GTAW (TIG) welding of Aluminium and Magnesium**!
Effect of Welding Parameters & Pulsed GTAW vs. GMAW	Increasing **Welding Current (I)** increases penetration depth; increasing **Arc Voltage (V ∝ arc length)** widens the weld bead; increasing **Welding Speed (s)** reduces heat input per unit length (H = V·I / s), narrowing both penetration and width. In **GTAW (TIG)**, the non-consumable tungsten electrode is alloyed with **1%–2% Thorium Oxide (ThO_2) or Cerium/Lanthanum Oxide** to raise thermionic electron emission and current capacity; **Pulsed-Current GTAW** alternates between a high Peak Current (melts spot) and a low Background Current (maintains arc while puddle cools, ideal for thin sheets and root passes). In thick pipe welding, **GTAW is used for the precision root pass** and **GMAW (MIG) is used for fast filler passes**!
Complete Lecture 14	
Orthogonal Cutting Geometry & 3 Heat Generation Zones	A cutting tool wedge has a **Rake Face** (inclined at **Rake Angle α**; positive rake reduces cutting force for ductile metals, while **negative rake angle** strengthens the cutting edge for hard steels and brittle carbide/ceramic inserts) and a **Flank Face** (inclined at **Relief/Clearance Angle γ** to prevent rubbing against the freshly machined surface). Heat is generated in **3 Zones**: (1) **Primary Shear Zone** along the shear plane (elasto-plastic deformation; **chips carry away 75%–80% of total cutting heat!**), (2) **Secondary Deformation Zone** at the tool-chip interface (sticking-sliding friction; 10%–20% heat into tool tip), and (3) **Tertiary / Flank Wear Zone** at the tool-workpiece interface (3%–5% heat).
4 Types of Metal Chips	(1) **Continuous Chip** (ductile metals like mild steel/Al cut at high speed, small feed, positive rake, good lubrication → best surface finish), (2) **Continuous Chip with Built-Up Edge (BUE)** (ductile metal at medium speed with high friction welds layers of work material onto the tool nose, which periodically break off and roughen the surface), (3) **Discontinuous (Segmental) Chip** (brittle metals like grey cast iron or bronze, or low cutting speed/large depth of cut → chip fractures into discrete segments), and (4) **Serrated (Saw-Tooth / Segmented) Chip** (semi-continuous chip with cyclic zones of high and low shear strain, typical of difficult-to-machine low-thermal-conductivity alloys like **Titanium and Austenitic Stainless Steel**).
Twist Drill Anatomy	**Twist Drill Anatomy:** Comprises *Shank, Body, Helical Flutes* (evacuate chips and deliver coolant), *Land, Margin* (narrow cylindrical guide strip), *Web/Core* (central backbone increasing in thickness toward shank), *Chisel Edge*, and two cutting *Lips* (standard point angle 118°). **Hole Operations Sequence: Drilling** originates a rough hole; **Boring** enlarges an existing drilled/cast hole using a single-point boring bar to correct eccentricity/axis alignment; **Reaming** removes a tiny margin (0.1–0.4 mm) with a multi-flute reamer for mirror surface finish and H7 dimensional tolerance; **Tapping** cuts internal threads (4-step cored hole

	sequence: *1. Cast core hole → 2. Chamfer entrance → 3. Drill to tap-drill diameter → 4. Cut internal threads with Tap*; external threads on rods are cut with a **Die** or single-point lathe threading tool). **Up Milling (Conventional) vs. Down Milling (Climb):** In *Up Milling*, cutter rotation opposes table feed (chip thickness starts at zero and grows to maximum; lifts workpiece upward). In *Down Milling*, cutter rotates in the same direction as table feed (chip starts at maximum thickness and exits at zero; presses workpiece down into fixture and gives better finish). *Milling Types:* **Slab (Peripheral) Milling** (teeth on cylinder periphery, axis parallel to surface), **Face Milling** (axis perpendicular to surface), and **End Milling** (cuts slots, pockets, and 3D contours). **Shaping vs. Planing & Centreless Grinding:** In *Shaping*, the single-point tool reciprocates horizontally (cutting on forward stroke, quick-return on backstroke) while the small workpiece feeds intermittently; in *Planing*, the massive workpiece reciprocates on a heavy bed table under a stationary tool. **Centreless Grinding** finishes cylindrical pins/shafts at high production rates without axial centers: the workpiece sits on an angled **Work Rest Blade** between a high-speed **Grinding Wheel** and a slow-speed rubber-bonded **Regulating Wheel** (tilted by a small feed angle α to control axial feed rate).
CNC Control Systems & Interpolation (Slide 24–26 of Cutting Deck)	Introduced in the early 1950s, a CNC system comprises **1. Data Input (G-code/M-code part program) → 2. Data Processing (Machine Control Unit MCU) → 3. Data Output (Servomotors + Ball-screws + Encoders)**. **Point-to-Point (PTP) Control:** Rapid positioning to exact (X, Y) coordinates before performing an operation at that spot (tool path between holes does not touch work; used in *CNC Drilling, Spot Welding, and Punching*). **Contouring (Continuous Path) Control:** Simultaneous coordinated velocity and position control of 2, 3, or 5 axes while the tool is continuously cutting (used in *CNC Milling, Turning, and Laser Cutting*) using **Linear Interpolation** (straight lines at any angle), **Circular Interpolation** (arcs/circles via G02/G03), or **Parabolic/Cubic Freeform Interpolation** (aerospace dies and turbine blades).
Complete Lecture 15	
Why AM (Cost vs. Complexity Curve)	In conventional subtractive/formative manufacturing, component cost rises exponentially with geometric complexity (custom dies, fixtures, multi-axis setups). In **Additive Manufacturing (AM)**, manufacturing cost is **virtually flat/independent of complexity** ("complexity is free"), enabling topology-optimized lightweight lattice structures, part consolidation (e.g., GE LEAP fuel nozzle consolidating 20 brazed parts into 1 printed part), zero tooling cost, and minimal material scrap!
6-Step AM Workflow & STL File Physics	**1. CAD Solid Model → 2. Conversion to .STL (STereoLithography / Standard Tessellation Language) → 3. Transfer to AM Machine & STL Manipulation (orientation, scaling, support generation, slicing) → 4. Machine Setup (layer thickness 0.05–0.3 mm, hatch strategy) → 5. Layer-by-Layer Build → 6. Removal & Post-Processing (support removal, UV/thermal curing, HIP, sanding)**. An `.STL` file approximates curved 3D surfaces using a mesh of **planar triangles**, each defined by the (X, Y, Z) coordinates of its 3 vertices plus an **outward-pointing unit normal vector** (using too few triangles causes faceted chordal error; thicker slicing layers cause the **Stair-Stepping Effect** on inclined surfaces).
All 7 ASTM	**All 7 ASTM F2792 Categories of Additive Manufacturing (Complete Slide Table — Mandatory 10-Mark Question!):**

ASTM Category	Feedstock Material Form	Energy / Bonding Source	Working Principle, Key Examples & Strengths/Limitations
1. Material Extrusion	Thermoplastic Filament (PLA, ABS, PETG, PEEK), Paste, or Concrete/Food	Resistive Heater in Liquefier Head	Pinch rollers feed filament into heated liquefier nozzle that extrudes roads layer-by-layer onto build platform. **Examples: FDM / FFF**. *Strengths:* Low cost, desktop-friendly. *Limitations (Slide 36):* Slow build speed, min layer thickness ~0.078 mm, nozzle radius prevents sharp inner corners, viscoelastic thermal warpage, and **Anisotropic strength (weakest along vertical Z-axis interlayer bonds!)**.
2. Vat Photopolymerization	Liquid Photopolymer Resin (Acrylates, Epoxies) in a Vat	UV Laser Beam or Digital DLP Projector	UV light selectively cures (crosslinks) liquid resin layer-by-layer as the build platform lowers/rises. **Examples: SLA (Stereolithography), DLP, CLIP**. *Strengths:* Highest surface finish & micron-level accuracy (jewelry, dental aligners). *Limitations:* Brittle photosensitive resins, requires support removal & post-UV curing.
3. Powder Bed Fusion (PBF)	Fine Polymer (Nylon PA12) or Metal (Ti6Al4V, Inconel, SS316L) Powder Bed	High-Power Laser or Electron Beam (in vacuum)	Roller spreads a thin powder layer; laser/electron beam selectively sinters or fully melts the cross-section. **Examples: SLS (polymers — surrounding unsintered powder acts as self-support!), SLM / DMLS, EBM (metals)**. *Strengths:* Fully dense functional aerospace/medical metal implants.
4. Binder Jetting	Powder Bed (Metal, Sand, Ceramic, Gypsum) + Liquid Binder	Piezoelectric/Thermal Inkjet Printhead + Furnace Sintering	Inkjet head selectively deposits liquid bonding agent onto powder bed at room temperature (no thermal distortion during printing!). **Examples: ExOne Metal Binder Jetting, Sand Mold Printing**. Green part requires post-print curing, debinding, and furnace sintering/infiltration.
5. Material Jetting	Liquid Photopolymer Droplets or Wax	Multi-Nozzle Piezo Inkjet Head + UV Curing Lamp	Sprays microscopic droplets of build & gel-support material like a 2D inkjet printer and cures each pass with UV light. **Examples: PolyJet, MultiJet**. *Strengths:* Multi-material & full-color anatomical models in a single build, ultra-smooth finish.
6. Directed Energy Deposition (DED)	Metal Powder Stream or Metal Wire blown/fed into Melt Pool	Focused Laser Beam, Electron Beam, or Plasma/Electric Arc	Coaxial nozzle feeds metal powder/wire directly into a moving focal melt pool on a 5-axis CNC head. **Examples: LENS (Laser Engineered Net Shaping), EBAM, WAAM**. *Unique Strength:* Can cladding/repair worn turbine blades and add features onto existing curved 3D metal parts!
7. Sheet Lamination	Rolls/Sheets of Paper, Polymer Film, or Metal Foil	Adhesive Roller + Laser Knife (LOM) or Ultrasonic Sonotrode (UAM)	Bonds successive sheets together and cuts each layer's perimeter contour using a CO_2 laser or CNC mill. **Examples: LOM (Laminated Object Manufacturing), UAM (Ultrasonic Additive Manufacturing)**. Fast for large solid blocks and embedded electronics.
The 3 Weld Zones (Exam Trap!)	(1) **Fusion Zone (Weld Metal):** Melted mixture of filler and base metal. (2) **Heat Affected Zone (HAZ):** Solid base metal right next to the weld pool that did NOT melt, but reached red-hot temperatures—causing coarse grain growth and making the **HAZ the mechanically WEAKEST zone most prone to cracking**! (3) **Unaffected Base Metal.**		

Arc Polarity	**DCEN (Direct Current Electrode Negative / Straight Polarity):** Electrons slam into the positive workpiece → **2/3 of arc heat is at the workpiece**, giving deep, narrow penetration. **DCEP (Electrode Positive / Reverse Polarity):** 2/3 heat at electrode → shallow weld with an **oxide-cleaning action** ideal for Aluminium/Magnesium!
SMAW (Shielded Metal Arc / Stick Welding)	Uses a consumable metal stick coated with flux. As the arc melts the stick, the flux vaporizes into a protective gas cloud and forms a liquid **slag blanket** over the cooling weld.
GTAW / TIG (Tungsten Inert Gas Welding)	Uses a **NON-CONSUMABLE Tungsten-thorium electrode** inside a torch that flows pure inert **Argon or Helium gas**. Filler metal is fed separately by the welder's other hand. Produces the **cleanest, spatter-free, highest-strength welds on thin sheets (<6 mm)** of stainless steel, Aluminium, and Titanium (slow, requires high skill).
GMAW / MIG (Metal Inert Gas Welding)	Uses a **CONSUMABLE bare wire electrode** fed continuously and automatically from a motor-driven spool through the welding gun, shielded by Argon + CO_2. One-handed, fast, high deposition rate for thick structural plates.
Soldering vs. Brazing vs. Welding	In **Soldering**, the base metals are **NEVER melted**! A low-melting filler alloy (Lead-Tin or Sn-Ag-Cu solder, melting point **< 450°C**; Brazing is > 450°C) is melted with a soldering iron and drawn into the microscopic gap between copper PCB pads and component leads by **Capillary Action** (after applying **Rosin/$ZnCl_2$ Flux** to dissolve surface oxides and improve wetting).

M4T4 Module 4, topic 4

Metal Cutting (Machining) & Introduction to CNC Machining

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	**Machining (Subtractive Manufacturing)** removes unwanted metal as chips using a hardened cutting tool (**Figure 4.2-2**):
Single-Point Tool — Engine Lathe Operations	On a Lathe, the **cylindrical workpiece rotates** in the Chuck while a single-point tool feeds against it: (1) **Turning:** Tool feeds parallel to axis to reduce outer diameter. (2) **Facing:** Tool feeds perpendicular to axis to machine a flat end face and reduce length. (3) **Taper Turning:** Feeds at an angle α to make a cone. (4) **Boring:** Enlarges an existing drilled hole internally. (5) **Knurling:** Embosses a diamond grip pattern.
Shaping vs. Planing	Both cut flat surfaces using a single-point tool in straight strokes, but in **Shaping** (small parts), the *cutting tool reciprocates* and the work is stationary; in **Planing** (huge machine beds), the *work table reciprocates* and the tool is stationary!
Multi-Point Tools — Hole Making & Milling	(1) **Drilling** creates a rough cylindrical hole; (2) **Reaming** lightly finishes an existing hole to exact tolerance and mirror smoothness; (3) **Tapping** cuts internal female screw threads inside a hole (*Die* cuts external male threads on a rod). In **Milling**, a rotating multi-tooth cutter removes metal: in *Up (Conventional) Milling*, cutter teeth rotate against the feed direction (chip thickness zero → max, tends to lift workpiece); in *Down (Climb) Milling*, teeth rotate with the feed (chip max → zero, presses work down onto table, better finish).
Introduction to CNC (Computer Numerical Control — Figure 4.2-2)	Replaces human handwheels with an automated closed-loop system: **[Part Program (G-codes for motion: G00 rapid, G01 linear, G02/G03 circular; M-codes for spindle/coolant)] → [Machine Control Unit (MCU)] → [Servo Amplifiers & Lead-Screw Motors] → [Machine Table/Spindle]** with real-time position & velocity feedback from an **Optical Rotary Encoder**! Two CNC motion control modes: (i) **Point-to-Point (PTP):** Only final X-Y coordinate matters, tool doesn't cut while moving between points (Drilling, Spot Welding); (ii) **Continuous Path**

	(Contouring): Tool cuts continuously along 2D/3D curves via Linear/Circular Interpolation (Milling, Lathe turning).

M4T5 Module 4, topic 5

Additive Manufacturing (3D Printing — Principles, .STL Pipeline & Applications)

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context	**Additive Manufacturing (AM / 3D Printing)** builds parts **layer-by-layer from the bottom up** directly from digital 3D CAD data with zero molds, dies, or fixtures (**Figure 4.2-3**):
The 5-Step 3D Printing Workflow	(1) Create a 3D solid model in CAD. (2) Export as an **.STL (Standard Tessellation Language / Stereolithography) file**, which approximates all outer surfaces using a mesh of connected 3D triangles. (3) Import into **Slicer Software** (Cura/PrusaSlicer) to slice the model into thin horizontal 2D layers (0.05–0.3 mm), generate temporary **support structures** for overhangs (>45°), and write G-code toolpaths. (4) Layer-by-layer build on the machine. (5) **Post-Processing** (dissolve/break supports, UV cure, sand, or hip-sinter).
The 7 ASTM	**The 7 ASTM Additive Manufacturing Categories (Highlight the Top 4 in FAT!):**
Material Extrusion (FDM — Fused Deposition Modeling)	A heated nozzle (200–250°C) melts and extrudes thermoplastic filament (**PLA, ABS, PETG, PEEK**) bead-by-bead. *Critical Exam Insight:* FDM parts are **Anisotropic**—strong along X-Y, but **30–50% weaker along the vertical Z-axis** because inter-layer bonding relies only on partial thermal diffusion and leaves microscopic voids!
Vat Photopolymerization (SLA — Stereolithography)	An ultraviolet (UV) laser traces each layer on the surface of a liquid photopolymer resin vat, solidifying it via photopolymerization (highest surface smoothness & micron detail: dental crowns, jewelry, hearing aids).
Powder Bed Fusion (SLS / SLM / DMLS)	A roller spreads a thin layer of Nylon (SLS) or Titanium/Inconel metal powder (SLM), and a high-power CO_2/fiber laser sinters or fully melts the cross-section. Unmelted powder in SLS naturally supports overhangs!
Directed Energy Deposition (DED / LENS)	Metal powder or wire is blown through a nozzle right into a laser melt pool—uniquely used to **repair cracked or worn aerospace turbine blades**! (Plus 5. *Binder Jetting*, 6. *Material Jetting / PolyJet*, and 7. *Sheet Lamination / LOM*).

M4T6 Module 4, topic 6

Fundamentals of Thermal Systems & Internal Combustion (IC) Engines

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context	In an **External Combustion (EC) Engine** (steam turbine), fuel burns outside in a boiler; in an **Internal Combustion (IC) Engine** (car/bike engine), fuel burns directly inside the cylinder on top of the piston (**Figure 4.3 Left**) (SMEC VIT Module 4 Notes).
IC Engine Geometry & Key Components	**TDC (Top Dead Center)** & **BDC (Bottom Dead Center)** = highest & lowest piston positions; **Stroke (L)** = distance from TDC to BDC; **Bore (D)** = cylinder inner diameter; **Swept Volume $V_s = (\pi/4)D^2L$**; **Clearance Volume V_c** = space above piston at TDC; **Compression Ratio $r = (V_s + V_c) / V_c$**. Key parts: *Piston Compression Rings* (seal gas) & *Oil Scraper Ring* (lubrication), *Gudgeon Pin*, *Connecting Rod* (converts reciprocating motion to rotary), *Crankshaft*, *Camshaft* (opens valves at half crank speed), and **Flywheel** (heavy cast-iron wheel that stores kinetic energy during the Power stroke and carries the piston smoothly through the 3 idle strokes!).

The 4 Strokes	**The 4 Strokes (720° = 2 Crankshaft Revolutions per Power Stroke, Figure 4.3-1):** (1) **Suction Stroke (TDC → BDC, Inlet Valve OPEN, Exhaust CLOSED):** Piston moves down, creating vacuum that draws air-fuel charge (SI) or pure air (CI). (2) **Compression Stroke (BDC → TDC, Both Valves CLOSED):** Piston moves up, compressing charge to high pressure and temperature. (3) **Power / Expansion Stroke (TDC → BDC, Both Valves CLOSED):** Spark plug fires (SI) or fuel injector sprays diesel into red-hot air (CI); high-pressure combustion gases slam piston down—**the ONLY stroke that produces positive work**! (4) **Exhaust Stroke (BDC → TDC, Exhaust Valve OPEN):** Piston pushes burnt gases out.
4-Stroke vs. 2-Stroke & SI (Petrol) vs. CI (Diesel)	A **2-Stroke Engine** completes the cycle in **1 crank revolution (360°)** using cylinder wall **Ports** (Inlet, Exhaust, Transfer) instead of valves—making it lighter with higher power-to-weight ratio, but *lower thermal efficiency and higher pollution* because some fresh fuel-air charge escapes directly out the open exhaust port during **Scavenging**! **SI (Petrol)** works on the constant-volume **Otto Cycle** (Spark plug, Carburetor/MPFI, low r = 6–10.5 to avoid knocking); **CI (Diesel)** works on the constant-pressure **Diesel Cycle** (Fuel injector, auto-ignition, high r = 14–22, higher thermal efficiency).

M4T7 Module 4, topic 7

Introduction to Power Plants (Thermal Steam, Nuclear, Hydro, Solar & Wind)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	Study the 3 classic Power Plant block diagrams in **Figure 4.3 (Right)**:
Coal-Fired Thermal Steam Power Plant (Rankine Cycle)	Operates on 4 circuits: (i) *Coal & Ash Circuit* (pulverized coal burns in boiler furnace; Electrostatic Precipitator ESP captures fly ash), (ii) *Air & Flue Gas Circuit* (Forced/Induced draft fans), (iii) *Feedwater & Steam Circuit*, and (iv) *Cooling Water Circuit* (Condenser + Cooling Tower). **3 Vital Heat-Recovery Accessories (FAT Favorites):** (a) **Superheater:** Heats wet boiler steam into dry superheated steam (prevents turbine blade erosion & boosts efficiency); (b) **Economizer:** Uses hot waste flue gas leaving the boiler to preheat incoming liquid feedwater—**saving 10–12% coal and boosting overall plant efficiency**! (c) **Air Preheater:** Uses remaining flue gas heat to warm combustion air before it enters the furnace.
Nuclear Fission Power Plant	Fission of **1 kg of Uranium-235 (U^{235}) releases as much heat as burning 4,500 tonnes of coal**! Inside the thick steel-and-concrete **Biological Shield** sits the Reactor Core containing: (a) **Fuel Rods:** Enriched UO_2 pellets in Zircaloy tubes; (b) **Moderator (Graphite, Heavy Water D_2O, or Light Water):** *Slows down* fast fission neutrons (2 MeV) to thermal speed (0.025 eV) so they can split more U^{235} nuclei; (c) **Control Rods (Boron or Cadmium):** Strong neutron-absorbing rods pushed into the core to slow/stop the chain reaction or pulled out to increase power! (In a *PWR — Pressurized Water Reactor*, primary water is kept at 155 bar so it never boils in the core, transferring heat via a Steam Generator; in a *BWR*, water boils directly in the core).
Hydroelectric Power Plant	Converts potential energy of water stored behind a **Dam/Reservoir** ($P = \rho \cdot g \cdot Q \cdot H \cdot \eta$) into kinetic energy through a high-pressure pipe called a **Penstock** (protected against sudden valve-closure "water hammer" pressure spikes by a vertical **Surge Tank**!). **Turbine Selection by Head H:** (i) **Pelton Wheel:** High Head (> 250 m), low flow (Impulse jet hitting bucket wheel); (ii) **Francis Turbine:** Medium Head (50–250 m, mixed-flow reaction); (iii) **Kaplan Turbine:** Low Head (< 50 m), high flow (axial propeller blades).

M4T8 Module 4, topic 8

Introduction to Refrigeration and Air Conditioning (VCRS, VARS & AC)

Overview & context

Refrigeration pumps heat *uphill* from a cold chamber to the warm room. Memorize the 4-component **Vapour Compression Refrigeration System (VCRS) diagram in Figure 4.4 (Left)**:

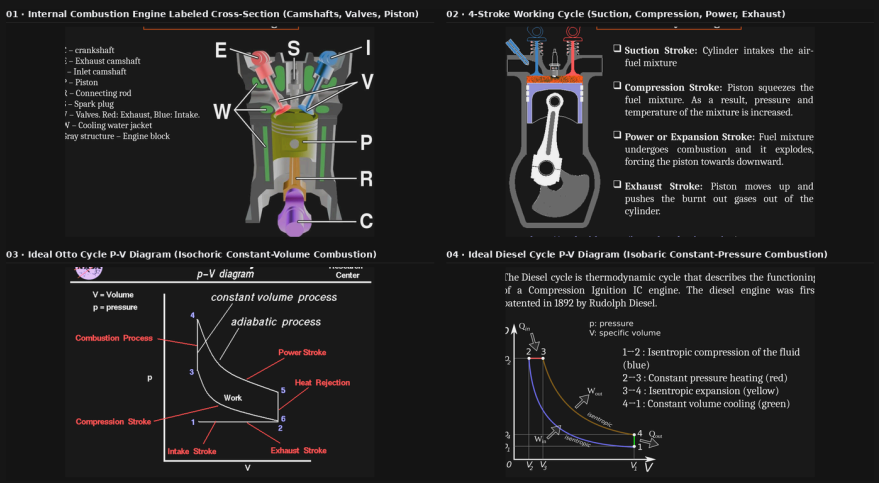


Figure 4.3B (Extracted from VIT Lecture 16 Slides): (a) IC Engine Labeled Cross-Section (E/I Camshafts, Valves, Piston, Connecting Rod, Crankshaft), (b) 4-Stroke Working Cycle, (c) Ideal Otto Cycle P-V Diagram (Constant-Volume Heat Addition), (d) Ideal Diesel Cycle P-V Diagram (Constant-Pressure Heat Addition).

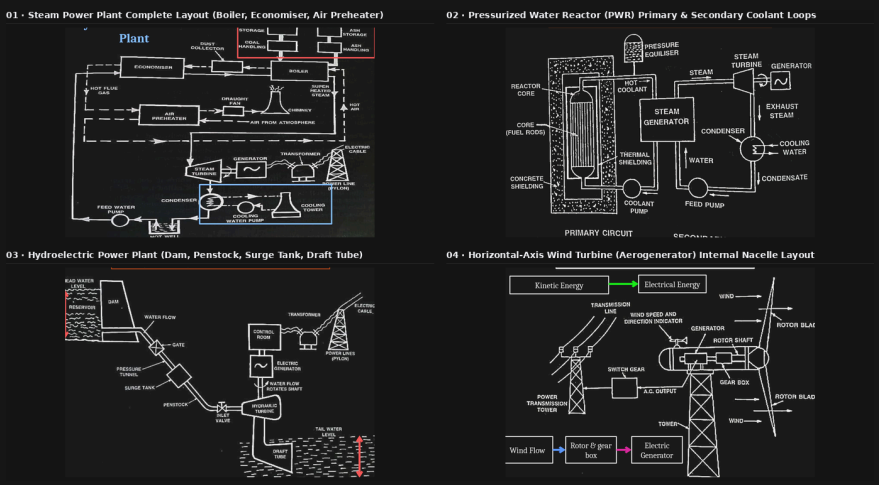


Figure 4.3C (Extracted from VIT Lectures 17 & 18 Slides): (a) Steam Power Plant Complete Layout (Economiser, Air Preheater, Superheater, Condenser), (b) Pressurized Water Reactor (PWR) Primary & Secondary Loops, (c) Hydroelectric Plant Layout (Penstock, Surge Tank, Draft Tube), (d) Horizontal-Axis Wind Turbine Layout.

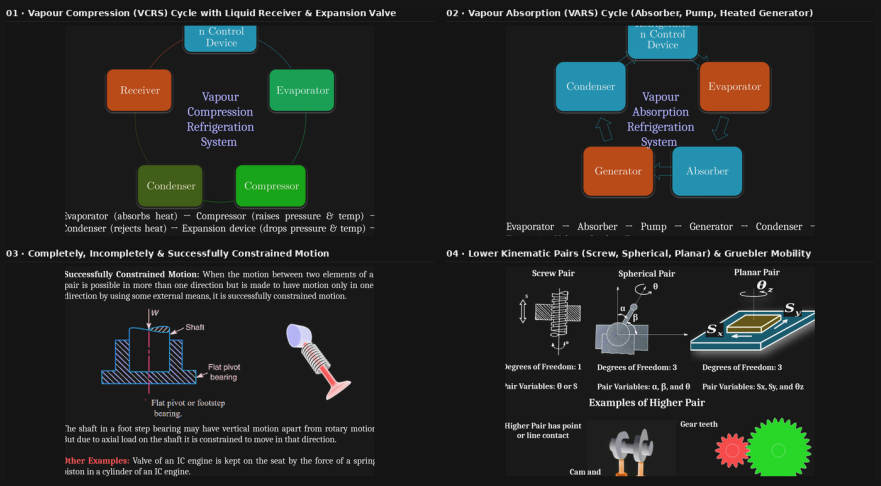


Figure 4.4B (Extracted from VIT Lectures 19 & 20 Slides): (a) Vapour Compression (VCRS) Cycle with Receiver, (b) Vapour Absorption (VARS) Cycle (Absorber, Pump, Generator), (c) Completely, Incompletely & Successfully Constrained Motion, (d) Lower Kinematic Pairs (Screw, Spherical, Planar).

Complete Lecture 16

Automobile Power Transmission Chain (Slide 2)	**A: IC Engine → B: Friction Clutch (operated by C: Clutch Pedal to engage/disengage engine shaft smoothly) → D: Manual/Automatic Gearbox → Propeller Shaft (with Universal Joints) → Differential Gear (splits torque 90° to rear/front axles) → Road Wheels**.
Detailed IC Engine Component Anatomy (Slides 4–11)	**Piston (Cast Al-Si Alloy):** Top *Crown/Head* receives explosive gas force; *Top Land & Ring Grooves* hold two upper **Compression Rings** (seal combustion gases and conduct 70% of piston heat into the water-cooled cylinder wall!) and one lower slotted **Oil Control (Scraper) Ring** (scrapes excess lubricating oil back to crankcase sump to prevent blue exhaust smoke); *Piston Boss* holds the hardened hollow **Piston Pin (Gudgeon/Wrist Pin)** retained by *Circlips*; lower *Piston Skirt* guides straight reciprocating motion. **Connecting Rod (Drop-Forged I-Beam Steel):** Small end fits piston pin; split **Big End** (clamped with cap, bearing inserts, bolts & castle nuts) fits the crank pin; converts reciprocating piston motion into rotary crankshaft motion under combined axial compression and whipping bending stress. **Crankshaft & Rim-Type Flywheel:** Crankshaft has *Main Journals* (supported in engine block bearings), *Crank Pin* (offset by crank radius r = Stroke L / 2), *Crank Webs*, *Counterweights* (balance reciprocating inertia), and drilled *Oil Passages*. The heavy cast-iron **Rim-Type Flywheel** concentrates mass at a large radius R to maximize moment of inertia ($I = mR^2$), storing excess kinetic energy during the 1 Power Stroke and releasing it during the 3 idle strokes (Suction, Compression, Exhaust) to minimize speed fluctuation!
Ideal Otto Cycle	**Ideal Otto Cycle (Constant-Volume Cycle) vs. Ideal Diesel Cycle (Constant-Pressure Cycle) P-V Diagrams:** **Otto Cycle (Petrol SI Engine): 0 → 1: Isobaric Intake** (air-fuel drawn in at atmospheric pressure); **1 → 2: Adiabatic (Isentropic, Q = 0) Compression** (piston rises from BDC to TDC, compressing mixture by compression ratio $r = V_1/V_2 = 8$ to 12); **2 → 3: Isochoric (Constant-Volume, ΔV = 0) Heat Addition Q_{in}** (spark plug fires at TDC; pressure jumps vertically from P_2 to P_3); **3 → 4: Adiabatic (Isentropic) Expansion Power Stroke** (high-pressure gas pushes piston to BDC, doing net work W_{out}); **4 → 1: Isochoric (Constant-Volume) Heat Rejection Q_{out}** (exhaust valve opens at BDC; pressure drops vertically); **1 → 0: Isobaric Exhaust**. Thermal efficiency: $\eta_{Otto} = 1 - 1 / r^{\gamma-1}$. **Diesel Cycle (Diesel CI Engine):** Only pure air is compressed during **1 → 2 Isentropic Compression** to a much higher compression ratio (r = 15 to 22, reaching ~600°C). Diesel

	fuel is sprayed continuously by the injector as the piston begins moving down from TDC, so combustion occurs as **2 → 3 Isobaric (Constant-Pressure, ΔP = 0) Heat Addition Q_{in}** (horizontal top line on the P-V diagram from V_2 to cut-off volume V_3!), followed by **3 → 4 Isentropic Expansion** and **4 → 1 Isochoric Heat Rejection**.
Complete Lectures 17	
Steam Power Plant Heat-Recovery Auxiliaries (Slide 7 of Lec 17)	Hot flue gases leaving the boiler pass sequentially through (1) **Superheater** (heats saturated steam above saturation temperature into dry superheated steam so turbine blades never suffer wet droplet erosion and Rankine efficiency increases), (2) **Economiser** (recovers waste flue-gas heat to preheat incoming boiler feedwater before it enters the boiler drum, saving 10%–15% coal!), and (3) **Air Preheater** (extracts remaining flue-gas heat before the chimney to preheat combustion air entering the furnace).
Nuclear Fission Equation	**Fission Reaction:** Bombarding Uranium-235 with a slow thermal neutron splits it into Barium and Krypton plus 3 neutrons and ~200 MeV of energy: $^{235}U_{92} + {}^{1}n_0 \rightarrow {}^{141}Ba_{56} + {}^{92}Kr_{36} + 3\,{}^{1}n_0$ **+ Energy**. Complete fission of just **1 kg of Uranium-235 releases heat equivalent to burning 4,500 tonnes of high-grade coal or 2,000 tonnes of oil**! **Reactor Core Elements: Moderator** (Graphite, Heavy Water D_2O, or ordinary H_2O slows down fast fission neutrons to thermal speeds so they can be captured by ^{235}U), **Control Rods** (**Boron or Cadmium** rods absorb excess neutrons to regulate the chain reaction at k = 1 or shut down the reactor), and **Biological Shielding** (inner steel thermal shield + thick outer concrete wall absorbs lethal gamma rays and neutrons). **PWR (Pressurized Water Reactor) vs. BWR (Boiling Water Reactor):** In a **BWR** (single-loop), water boils directly inside the reactor core into radioactive steam that drives the turbine. In a **PWR** (two-loop, shown in Figure 4.3C-b), a **Pressurizer (Pressure Equalizer)** keeps the primary reactor coolant water under high pressure (~155 bar) so it reaches 315°C *without boiling*; this radioactive primary water flows through a closed **Steam Generator (Heat Exchanger)** to boil completely separate, non-radioactive secondary water into steam for the turbine!
Hydroelectric Power Plant	**Why the Draft Tube (Component #7) is Essential on Reaction Turbines:** Connected between the Francis/Kaplan turbine runner outlet and the tailrace, the **Draft Tube** is a diverging pipe (increasing cross-sectional area) that **converts the high leftover kinetic energy ($V^2/2g$) of the exiting water into useful static pressure energy**, creating a negative suction head at the runner exit so the turbine can be mounted safely above tailrace flood level without losing net head! (Note: Hydro turbines run at low speeds of **300–400 rpm** and hydro plants last **100–125 years**). **Turbine Selection by Head & Specific Speed (Slide 7 Graph):** (1) **Pelton Wheel (Impulse Turbine): High Head (> 250 m), Low Flow, Low Specific Speed** — nozzle converts all head into a high-velocity free water jet striking double-hemispherical buckets tangentially at atmospheric pressure. (2) **Francis Turbine (Mixed-Flow Reaction): Medium Head (40 – 250 m), Medium Flow & Specific Speed** — water enters radially inward through adjustable guide vanes and exits axially through a draft tube. (3) **Kaplan Turbine (Axial-Flow Propeller Reaction): Low Head (< 40 m), High Flow, High Specific Speed** — water flows parallel to the shaft axis across adjustable-pitch propeller blades.
Solar PV vs	**Solar PV vs. Concentrated Solar Power (CSP) vs. Solar Thermal (Slides 11–16 of Lec 18):** **Solar Photovoltaic (PV):** Semiconductor PN solar cells convert sunlight photons **directly into DC electricity** via the Photovoltaic Effect (**Solar Panel → Charge Controller / Battery → DC-to-AC Inverter → AC Grid/Load**). **Concentrated Solar Power (CSP):** Curved **Parabolic Trough Mirrors or Heliostat Towers** focus sunlight onto a receiver tube carrying heat-transfer fluid → produces high-pressure steam in a **Heat Exchanger → drives a Steam Turbine & AC Generator**!

	Solar Thermal Heating: Rooftop flat-plate or evacuated-tube collector absorbs solar heat into a working fluid loop → heats domestic water in an insulated storage tank.
Complete Lectures 19	
Vapour Compression (VCRS)	In the Lecture 19 VCRS diagram (Figure 4.4B-a), a liquid **Receiver (storage reservoir)** is placed right between the Condenser outlet and the Expansion Control Device (Capillary tube / Thermostatic Expansion Valve) to store condensed high-pressure liquid refrigerant (commonly **R-134a** in automotive/domestic refrigerators and **R-410A / R-32** in split air conditioners). In the **Vapour Absorption Refrigeration System (VARS, Figure 4.4B-b)**, the electricity-hungry mechanical compressor is replaced by an **Absorber → Liquid Solution Pump → Heated Generator (Boiler) → Pressure-Reducing Valve** loop: low-pressure ammonia (NH_3) vapour leaving the Evaporator is absorbed by cold weak water in the **Absorber** (rejecting heat to cooling water); a tiny **Liquid Pump** pumps the strong Aqua-Ammonia solution to high pressure into the **Generator**, where low-grade heat (solar, waste steam, or LPG flame) boils off high-pressure NH_3 vapour into the Condenser while weak water returns to the Absorber via a pressure-reducing valve!
10-Point Comparison	1. *Purpose:* Preserves food/medicine inside a small insulated cabinet vs. controls temperature, humidity, and air purity of an occupied room/building. 2. *Target Temperature Range:* **0°C to 5°C** in fresh food cabinet and **−18°C** in freezer vs. **16°C to 30°C** human comfort zone (typically 22°C–25°C). 3. *Volume Cooled:* Small sealed volume (150–600 Liters) vs. large open room volume (30–100+ m^3). 4. *Evaporator & Condenser Placement:* Both mounted on the same cabinet (evaporator inside freezer, condenser coil on back/sides rejecting heat into the kitchen!) vs. physically separated across the building wall (evaporator indoor unit inside room, condenser outdoor unit outside building!). 5. *Air Circulation & Compressor Capacity:* Natural convection or tiny internal fan with fractional-HP hermetic compressor (0.1–0.25 TR) vs. high-CFM centrifugal blower and heavy-duty rotary/scroll compressor (1.0 TR, 1.5 TR, or 2.0 TR).
Lecture 20 Slide	**Lecture 20 Slide Notes — Constrained Motion, Kinematic Links/Pairs & Gruebler's 5-Bar Proof:** **3 Types of Constrained Motion (Slides 5–8, Figure 4.4B-c):** 1. **Completely Constrained Motion:** Relative motion between a pair is limited to a single definite direction *purely by the geometry of the pair itself*, regardless of the direction of external force. *Slide Examples:* A **square bar sliding in a square hole**, or a **circular shaft with thrust collars at both ends rotating in a circular hole**. 2. **Incompletely Constrained Motion:** Relative motion can take place in more than one independent direction depending on the applied force. *Slide Example:* A **smooth circular shaft without collars inside a circular hole** (can both rotate θ and slide axially S independently — 2 DOF!). 3. **Successfully (Partially) Constrained Motion:** Geometry alone allows more than one motion, but an *external force or spring* prevents the unwanted motion so it behaves as 1-DOF in service! *Slide Examples:* A **vertical shaft rotating in a Footstep Bearing** (downward gravity load W prevents upward axial lift), an **IC Engine Cam-Operated Poppet Valve** (valve spring keeps follower in contact with cam), and a **Piston reciprocating inside an Engine Cylinder** (gas pressure and crank geometry prevent rotation). **Complete 6 Lower Kinematic Pairs Table (Slide 14–15):** (1) **Revolute / Turning Pair (R):** 1 DOF (pure rotation θ; pin/hinge joint), (2) **Prismatic / Sliding Pair (P):** 1 DOF (pure translation S; piston in cylinder / crosshead), (3) **Screw / Helical Pair (H): 1 DOF** (rotation θ and translation S are coupled by lead L: $S = L \cdot \theta/(2\pi)$; bolt and nut / lathe leadscrew), (4)

	Cylindrical Pair (C): 2 DOF (independent θ and S; uncollared shaft in hole), (5) **Spherical / Globular Pair (G): 3 DOF** (3 rotations α, β, θ; ball-and-socket car mirror / joystick), and (6) **Planar Pair (E): 3 DOF** (2 translations S_x, S_y + 1 rotation θ_z; flat block resting on a flat surface). **Gruebler's Planar Mobility Proof for Slider-Crank (4-Bar) vs. 5-Bar Mechanism (Slides 17–19):** Using **$F = 3(N - 1) - 2P_1 - 1P_2$**: *4-Bar Linkage / Slider-Crank:* N = 4 links, P_1 = 4 lower pairs, P_2 = 0 ⇒ **$F = 3(4 - 1) - 2(4) = 9 - 8 = 1$ DOF** (single input motor completely defines the mechanism's motion!). *Pentagonal 5-Bar Linkage (Slide 19):* N = 5 links (1 ground link + 4 moving links), P_1 = 5 revolute pin joints, P_2 = 0 ⇒ **$F = 3(5 - 1) - 2(5) = 12 - 10 = 2$ DOF**! Because F = 2, a 5-bar linkage requires **two independent input actuators** (e.g., two servo motors driving the two base cranks) to produce a constrained 2D programmable robot end-effector trajectory!
The 4 VCRS Components in Order (1 → 2 → 3 → 4)	(1) **Evaporator (Freezer Coils inside fridge):** Cold low-pressure liquid refrigerant (R-134a / R-32 / R-600a) absorbs latent heat **Q_L** from your food/room and boils into low-pressure vapor. (2) **Compressor (Heart of the system):** Driven by an electric motor (work input **W_{in}**), it compresses the vapor to high pressure and high temperature (above room temp!). (3) **Condenser (Black coils on back of fridge / outdoor AC unit):** Hot high-pressure vapor dumps heat **$Q_H = Q_L + W_{in}$** to outside air and condenses into high-pressure liquid. (4) **Expansion Valve / Capillary Tube:** A narrow throttling restriction that drops the pressure and temperature of the liquid drastically back to freezing cold (**isenthalpic throttling, $h_3 = h_4$**) before re-entering the Evaporator!
COP & 1 Ton of Refrigeration (TR)	Performance is measured by **Coefficient of Performance: COP = Desired Cooling Effect (Q_L) / Work Input (W_{in})** (typically 3.0 to 4.5). **1 Ton of Refrigeration (1 TR) = 3.517 kW (≈ 3.5 kW or 210 kJ/min)**—defined as the rate of heat extraction needed to freeze 1 short ton (1,000 kg) of liquid water at 0°C into solid ice at 0°C in 24 hours.
Vapour Absorption System (VARS) & Air Conditioning (AC)	**VARS** replaces the electricity-hungry mechanical compressor with a heat-driven **[Absorber + Liquid Pump + Generator/Boiler]** using Ammonia-Water (NH_3 refrigerant, H_2O absorbent) or Lithium Bromide-Water (H_2O refrigerant, LiBr absorbent), running on waste factory steam or solar heat! An **Air Conditioner** uses a VCRS loop + air filter + blower fan to simultaneously control 4 human comfort parameters: **Temperature (22–25°C), Relative Humidity (40–60% RH via dehumidification), Air Purity, and Air Velocity**.

M4T9 Module 4, topic 9

Basics of Fuel Cells (Hydrogen PEMFC) & Sustainable Energy Technologies

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	A **Fuel Cell** is an electrochemical device that converts the chemical energy of a continuous fuel (**Hydrogen H_2**) and oxidant (**Oxygen O_2 from air**) directly into **DC Electricity + Water + Heat** without combustion (bypassing the Carnot heat-engine bottleneck to achieve **45–60% electrical efficiency**!).
Proton Exchange Membrane	
Anode (− Electrode, Platinum Catalyst)	Hydrogen gas splits into positive protons and free electrons: **$H_2 \rightarrow 2H^+ + 2e^-$**.
Polymer Electrolyte Membrane (PEM)	A solid fluoropolymer membrane that allows **ONLY positive protons (H^+)** to pass directly through to the cathode while **blocking electrons (e^-)**—forcing the electrons to travel

through the external wire to power the EV motor or load!

Cathode (+ Electrode)	Arriving protons and external-circuit electrons combine with oxygen from air to produce pure water: $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$ **(Pure Water!)**.
Solar PV vs. Solar Thermal (CSP), Wind & Biomass	**Solar Photovoltaic (PV)** cells use doped Silicon PN junctions to convert sunlight photons directly into DC electricity via the Photovoltaic Effect (then a Power Electronic **Inverter** converts DC to 230 V AC). **Concentrated Solar Power (CSP)** uses mirrors (parabolic troughs/heliostats) to focus sunlight onto a receiver tube to boil steam for a turbine. **Wind Turbines** convert wind kinetic energy ($P = \frac{1}{2} \rho A v^3 \cdot C_p$, where wind power scales with the **CUBE of wind speed v^3**; Betz limit $C_{p,max}$ = 59.3%) via rotor blades, a step-up gearbox, and a generator inside the Nacelle.

M4T10 Module 4, topic 10

Introduction to Mechanisms (Four-Bar Chain, Inversions & Stewart Hexapod)

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	A **Kinematic Link** is a rigid body; a **Kinematic Pair** is a joint between two links; a **Kinematic Chain** is a closed loop of links yielding constrained motion (1 input gives 1 definite output); and fixing one link of a kinematic chain creates a **Mechanism** (**Figure 4.5 Panels 1 & 2**) (SMEC VIT Module 4 Notes).

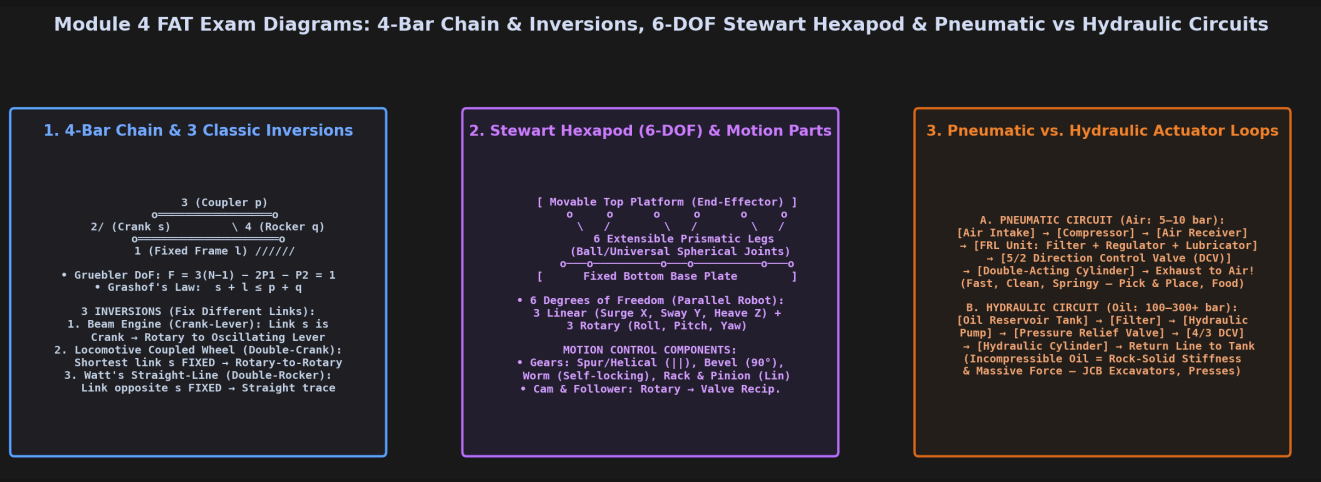


Figure 4.5: FAT Exam Diagrams for Topic 10 (4-Bar Inversions & 6-DOF Hexapod), Topic 11 (Motion Parts), and Topic 12 (Pneumatic/Hydraulic Circuits).

Lower Pair vs. Higher Pair & Gruebler's DoF Criterion	*Lower Pairs (P_1)* have **surface/area contact** (Revolute/Pin joint, Prismatic/Sliding piston, Screw, Cylindrical, Spherical ball-and-socket). *Higher Pairs (P_2)* have **point or line contact** (Cam & Follower, meshing Gear teeth, ball bearings). Planar Degrees of Freedom (Mobility F): $\mathbf{F = 3(N - 1) - 2P_1 - P_2}$. For a 4-bar linkage (N = 4 links, P_1 = 4 pin joints, P_2 = 0): $\mathbf{F = 3(3) - 2(4) = 1}$ **DOF**.
Grashof's Law &	**Grashof's Law & The 3 Classic Inversions of a Four-Bar Chain (Figure 4.5-1):** Grashof's Law states that at least one link can revolve a full 360° if $\mathbf{s + l \leq p + q}$ (shortest + longest ≤ sum of other two). By fixing different links of a 4-bar chain (**Kinematic Inversion**), we get:
Inversion 1 — Beam Engine (Crank-and-Lever / Crank-Rocker)	Link adjacent to shortest link is fixed; shortest link *s* rotates 360° (Crank) while opposite link oscillates (Lever) to pump water or drive a steam piston.

Inversion 2 — Coupled Wheels of a Locomotive (Double-Crank)	Shortest link *s* is fixed as the frame and opposite links are equal in length—both links adjacent to the frame rotate 360° in tandem to couple train drive wheels!
Inversion 3 —	**Inversion 3 — Watt's Straight-Line Indicator / Ackermann Steering (Double-Rocker):** Link opposite the shortest link is fixed—both links hinged to the frame only oscillate (rockers), and the midpoint of the coupler traces an accurate straight line!
Hexapod Mechanism (6-DOF Stewart Platform — Figure 4.5-2)	A **Parallel Manipulator** consisting of a fixed bottom base plate, a movable top platform (end-effector), and **6 independent extensible linear actuators (hydraulic/electric prismatic legs)** attached via universal/spherical ball joints. By varying the 6 leg lengths simultaneously, the top platform achieves all **6 Spatial Degrees of Freedom**: **3 Linear Translations (Surge X, Sway Y, Heave Z) + 3 Rotations (Roll, Pitch, Yaw)**. Because all 6 legs share the load in parallel, it has vastly higher stiffness, accuracy, and payload capacity than a serial robot arm (used in **pilot flight simulators, 6-axis CNC machining, surgical robots, and giant telescope mirrors**).

M4T11 Module 4, topic 11

Mechanical Components & Motion Control in Mechanical Systems

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Gear Drives (Positive, Zero-Slip Tooth Mesh)	(1) **Spur Gears:** Straight teeth parallel to shaft axis—connects parallel shafts at moderate speeds (noisy at high RPM due to sudden line tooth impact). (2) **Helical Gears:** Teeth cut at a helix angle—gradual point-to-line engagement gives silent, smooth, high-speed transmission between parallel shafts (car gearboxes). (3) **Bevel Gears:** Conical gears connecting **intersecting shafts at 90°** (automobile differential). (4) **Worm & Worm Wheel:** Screw-like Worm meshes with a wheel between **non-intersecting 90° shafts**—gives huge single-stage speed reduction (20:1 to 100:1) and **self-locking** (hoists/cranes cannot fall backward!). (5) **Rack & Pinion:** Circular pinion gear meshes with a flat toothed bar (Rack) to convert **Rotary ↔ Linear motion** (car steering & lathe carriage).
Cam & Follower, Belt/Chain Drives, Bearings, Clutches & Brakes	A rotating **Cam** (non-circular lobe) pushes a **Follower** (Knife-edge, Roller, or Flat-faced) to convert continuous rotation into precise timed reciprocating motion (opening/closing IC engine inlet & exhaust valves!). **Belt Drives** (Flat, V-belt, Toothed Timing belt) transmit power between distant shafts quietly with overload slip protection; **Chain & Sprocket Drives** give zero-slip positive transmission (bicycles/motorcycles). **Bearings** support rotating shafts with low friction: *Sliding Contact (Journal/Sleeve bearing)* uses a pressurized oil film for heavy turbines; *Rolling Contact (Ball & Roller bearings)* uses hardened steel balls/rollers for low starting friction.

M4T12 Module 4, topic 12

Introduction to Robotics & Automation (Pneumatic vs. Hydraulic Actuators)

Concepts & Working ✓ Key Equations Slide Diagrams Solved PYQ

Overview & context

An industrial robot uses **Sensors** (eyes/ears), a **Controller** (brain), and **Actuators** (muscles: Electric, Pneumatic, or Hydraulic) to manipulate an **End-Effector / Gripper**. Compare the **Pneumatic and Hydraulic Actuator Circuits in Figure 4.5 (Panel 3)**:

Parameter	Pneumatic Actuator System (Compressed Air)	Hydraulic Actuator System (Pressurized Oil)
Working Fluid & Pressure	**Compressed Air** at low pressure (**5 to 10 bar**).	**Pressurized Mineral Oil** at very high pressure (**100 to 300+ bar**).

Parameter	Pneumatic Actuator System (Compressed Air)	Hydraulic Actuator System (Pressurized Oil)
Compressibility & Stiffness	Air is **highly compressible (springy/spongy)**—hard to hold exact mid-stroke positions under varying loads.	Liquid oil is **virtually INCOMPRESSIBLE**—gives rock-solid stiffness and holds heavy loads mid-air without drifting!
Circuit Loop & Components (Figure 4.5-3)	**Open Loop:** Compressor → Air Receiver → **FRL Unit (Filter, Regulator, Lubricator)** → 5/2 DCV → Cylinder → **Exhausts directly to atmosphere** (no return pipe!).	**Closed Loop:** Oil Tank Reservoir → Filter → Positive-Displacement Pump → **Pressure Relief Valve** → 4/3 DCV → Cylinder → **Return Line back to Tank**.
Speed, Force, Cleanliness & Applications	**Ultra-fast speed**, 100% clean (no oil fire/contamination risk), low/medium force → **High-speed pick-and-place robots, food/pharma packaging, bus doors, dental drills**.	**Massive force & torque** (25× pneumatic), moderate speed, risk of messy oil leaks → **JCB excavators, heavy forging presses, aircraft landing gear, heavy foundry robots**.

M4T13 Module 4, topic 13

Module 4 FAT PYQ Self-Assessment & Multi-Concept Solved Problems

Concepts & Working ✓ | Key Equations | Slide Diagrams | Solved PYQ

Overview & context	Test your mastery on these comprehensive VIT Module 4 questions (SMEC VIT Module 4 Notes & Question Bank).

PYQ 1 [10 Marks — Multi-Concept Manufacturing Selection]: Casting + Forming + Welding + 3D Printing

Overview & context	**Question:** An automotive company is manufacturing four parts for a new hybrid vehicle: (i) A hollow cast-iron engine cylinder block with complex internal water jackets, (ii) A high-fatigue alloy-steel crankshaft, (iii) A thin 2 mm aluminium battery enclosure requiring leak-proof joints, and (iv) A custom patient-matched lattice titanium bracket (batch size = 1). Select the exact manufacturing process for each, sketch the mold/setup for (i) and (iii), and justify your choice.
(i) Engine Cylinder	Sketch Figure 4.1(1). Cast iron has high fluidity and low shrinkage; complex internal cooling passages are formed using baked **Dry Sand Cores** supported by **Chaplets** and fed by an open **Riser**.
(ii) Alloy-Steel Crankshaft → Closed-Die Hot Forging (2 Marks)	Sketch Figure 4.1(2B). Forgingabove recrystallization temperature aligns metal crystal grains continuously along the crank web contour (**unbroken grain flow**), giving unmatched fatigue and impact resistance that casting or machining from bar stock cannot match.
(iii) Thin 2	**(iii) Thin 2 mm Aluminium Battery Enclosure → GTAW / TIG Welding with DCEP/AC & Pure Argon (3 Marks):** Sketch Figure 4.2(1). Thin Al sheets burn through easily and form a tough Al_2O_3 oxide film; **TIG welding** with a non-consumable Tungsten electrode and pure Argon gas shield provides precise heat control, oxide cleaning, and zero spatter.
(iv) Custom Lattice	Sketch Figure 4.2(3). Exporting the CAD model to an **.STL triangle mesh** and laser-melting titanium powder layer-by-layer requires zero tooling cost and builds internal hollow lattices impossible to machine via CNC.

PYQ 2 [10 Marks — Multi-Concept Thermal & Energy]: Power Plant Accessories + VCRS Numerical + PEM Fuel Cell

Overview & context	**Question:** (a) Explain the specific function of the Economizer in a Steam Power Plant, the Moderator and Control Rods in a Nuclear Reactor, and the Surge Tank in a Hydroelectric Plant. (b) A domestic refrigerator working on the VCRS cycle extracts 1.5 TR of heat from a cold chamber while its compressor consumes 1.25 kW of electrical power. Calculate the heat rejected at the condenser (in kW) and the COP of the refrigerator.

Part (a) Power Plant Critical Components (6 Marks)	(1) **Economizer (Steam Plant, Figure 4.3-2A):** Captures waste heat from hot flue gases leaving the boiler to preheat incoming feedwater before it enters the boiler drum—saving 10–12% fuel and preventing thermal shock to the boiler. (2) **Moderator vs. Control Rods (Nuclear Plant, Figure 4.3-2B):** The *Moderator* (Heavy Water D_2O or Graphite) slows down fast fission neutrons (2 MeV → 0.025 eV thermal neutrons) to sustain U^{235} fission; *Control Rods* (Boron or Cadmium) absorb excess neutrons to regulate reactor power or shut it down in an emergency. (3) **Surge Tank (Hydro Plant, Figure 4.3-2C):** A vertical open standpipe connected to the Penstock that absorbs violent high-pressure shockwaves (**Water Hammer**) when turbine gates close suddenly.
Part (b) VCRS Numerical (4 Marks)	Cooling load in kW: Since **1 TR = 3.517 kW (≈ 3.5 kW)**, refrigerating effect $\mathbf{Q_L = 1.5 \times 3.5 = 5.25\ kW}$. Compressor work input: $\mathbf{W_{in} = 1.25\ kW}$. Coefficient of Performance: $\mathbf{COP = Q_L / W_{in} = 5.25 / 1.25 = 4.20}$. By Conservation of Energy across the VCRS loop (Figure 4.4-1), heat rejected at the Condenser is: $\mathbf{Q_H = Q_L + W_{in} = 5.25 + 1.25 = 6.50\ kW}$.

PYQ 3 [10 Marks — Mechanisms & Robotics]: Grashof's Law, DoF Numerical, Stewart Hexapod & Actuators

Overview & context	**Question:** (a) A planar four-bar linkage has link lengths of 40 mm, 100 mm, 70 mm, and 80 mm. Verify whether it satisfies Grashof's Law, compute its Degrees of Freedom using Gruebler's criterion, and state the type of mechanism obtained when the 40 mm link is fixed vs. when a link adjacent to the 40 mm link is fixed. (b) Sketch and compare a 6-DOF Stewart Platform Hexapod and explain why Hydraulic actuators are chosen over Pneumatic actuators for heavy aircraft flight simulators.
Part (a) Grashof & Gruebler Numerical (5 Marks)	Shortest link **s = 40 mm**, longest link **l = 100 mm**, other links **p = 70 mm, q = 80 mm**. Check Grashof's condition: **s + l = 40 + 100 = 140 mm**, and **p + q = 70 + 80 = 150 mm**. Since **s + l (140 mm) ≤ p + q (150 mm)**, it **satisfies Grashof's Law**! Degrees of Freedom (N = 4 links, P_1 = 4 revolute pin joints, P_2 = 0): $\mathbf{F = 3(N - 1) - 2P_1 - P_2 = 3(4 - 1) - 2(4) = 9 - 8 = 1\ DOF}$. (i) When shortest link (40 mm) is fixed → **Double-Crank (Drag-Link / Locomotive Coupled Wheel) mechanism** (both adjacent links rotate 360°). (ii) When an adjacent link (70 mm or 80 mm) is fixed → **Crank-Rocker (Beam Engine) mechanism** (40 mm link rotates 360° while opposite link oscillates).
Part (b) Stewart Platform & Hydraulic vs. Pneumatic (5 Marks)	Sketch Figure 4.5(2 & 3). The **Stewart Platform Hexapod** uses 6 parallel extensible legs between a fixed base and a top platform to control 3 translations (Surge, Sway, Heave) and 3 rotations (Roll, Pitch, Yaw). Heavy flight simulators use **Hydraulic Actuators (pressurized incompressible oil at 100–300 bar)** rather than Pneumatic actuators because oil is incompressible—providing massive force capacity, instant high-stiffness positioning, and zero spongy bounce under multi-ton cabin loads!